INTERVIEWEE: **Ronald D. Macfarlane**

INTERVIEWER: **Michael A. Grayson**

LOCATION: **Texas A&M University, College Station, Texas**

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GRAYSON: So, I'm going to start the way we usually do by saying, my name is Mike Grayson, and I'm sitting in the office of Professor Ron Macfarlane in Texas A&M University in College Station, Texas. Today is 26 May 2011. A nice pleasant day, but it'll get warm before everything's over. So, we're going to do a little oral history interview of Ron's career. Normally, we like to start these back in the beginning of time, so do we have information on your birthdate and location? I don't think so.

MACFARLANE: No.

GRAYSON: So when were you born?

MACFARLANE: Nineteen thirty-three, 21 February, in Buffalo, New York.

GRAYSON: Okay. So you're a real transplant to this part of the country.

MACFARLANE: That's right.

GRAYSON: And your parents' names?

MACFARLANE: William Arthur MacFarlane and Margaret Ann Baker.

GRAYSON: Okay. Were there any other children in the family, or you were . . . ?

MACFARLANE: Yes. I was the oldest. I have a brother and a sister.

MACFARLANE: [Yes].

GRAYSON: Who wants to do that?

MACFARLANE: They will. They love to surf the internet and I take advantage of that. So when they find a hot website, that's exactly what we're talking about, they'll share it with me.

GRAYSON: Oh, good.

MACFARLANE: Then I'll scan the website and I could email that website to the class. So it's very dynamic. The discovery part of science. They can get that feeling for did we discover something that's relevant. That's why I feel now my teaching and my research, it's the same mindset to come up with new ideas, thinking out of the box about the . . .

GRAYSON: Now you teach two sections?

MACFARLANE: I only teach one section. Another person teaches another section, which I'd say the traditional way you know, a textbook and . . .

GRAYSON: But then quantitative analysis has a lab associated with it too, doesn't it?

MACFARLANE: [Yes]. And that's something that I'm working at, and the lady that runs that, she knows what I'm doing. She's heard feedback from them. It's not traditionally . . . it's been treated like a totally separate course.

GRAYSON: The lab course?

MACFARLANE: I see the virtue of having it linked with my lectures. Now for example, she knows that there's one example, that Kjeldahl nitrogen analysis. I use the Kjeldahl analysis as a paradigm for titration. You can get into what the nitrogen analysis involves. Is it connected with proteins? And the proteins are something that's important for Third World nutrition, and then we're continuing and developing new agriculture products that are high in protein. That's a big deal, is the protein analysis. Then we're actually making beef that's been raised on a particular diet that's supposed to be rich in protein and low in fat. I don't know where I'm going on that. But then . . . okay, so I lost my train of thought on that one.

GRAYSON: Well, using Kjeldahl as a model for—

MACFARLANE: [Yes], right. [. . .] Right. But then the other things like the marijuana detection in urine, and analysis of the sunscreen, and there was a really neat application of Beer's Law using the sun as a light source to measure the ozone content in the upper atmosphere. The students found an article on that, and they made their presentation on the use of the Beer's Law.

GRAYSON: That's neat.

MACFARLANE: [Yes]. I would never have thought of that, but they have a whole collection now of really interesting PowerPoint presentations.

GRAYSON: You know, the idea that the subject that you're learning is connected in many ways to the real world, and they're giving you real-time information about the world, just makes it a heck of a lot better than just . . .

MACFARLANE: Well, that's the word I get from the students. "We're willing to do that, if you can relate to our lives". And I remember one time I had a student come to me, it was very close to the beginning of class last semester. He had failed the first exam. Said, "I've never failed an exam in my life and this could be devastating to me, because I want to go to med school."

GRAYSON: Oh, yes, the old med school problem.

MACFARLANE: And so I said, "Well, just show me how you're studying." He said—and it was obvious he was memorizing. I said, "That's your problem." He said, "What? There's no other way." I said, "Yes, conceptual understanding." "What's that mean?" I gave him an example. "I never thought of that." I noticed that all the rest of the semester he got As, but he never, he never came back to me to visit me here <**T: 150 min**> except six years later, he came back to my office—

GRAYSON: So this is six years ago?

MACFARLANE: Six years later.

GRAYSON: Later, [yes].

MACFARLANE: And he said, "Remember me?" I said, "Vaguely." He said, "Well, I came to your office and I failed my exam. I was upset about it. You told me what conceptual learning is." I said, "Oh, I remember now." "Well, today I graduated from med school, and you are the reason I made it." Well, that made a good day.

GRAYSON: [Yes].

MACFARLANE: Because I learned what conceptual learning means, because conceptual understanding of theory comes . . . [fingers snapping]

GRAYSON: [Yes]. [Yes], and besides, I think you would want your MD to learn that way.

MACFARLANE: That's right. Well, they do actually, because my wife has been through that now. And the medical school's actually realized this. In fact, the first complaint they get from their students is they're memorizers. It's something; you cannot be a memorizer and be an MD. The way that the medical school has developed [now] is there is lots of shadowing and hands-on stuff. So that's another way of getting reality to us. So you spend years in med school and as a resident to get practical experience, but that's another way of . . .

GRAYSON: Definitely, [yes].

MACFARLANE: [Yes]. So now, I'm amazed that that stuff my wife comes up with. Oh, but some little detail. "How do you remember that?" "Well, we learned that in med school; when we see some patients that have that, it kind of sinks in."

GRAYSON: [Yes]. I'm sure they've seen a lot of patients with that condition.

MACFARLANE: So medical research is very, very well designed and works really well.

GRAYSON: What do you think about a break for lunch?

MACFARLANE: Okay. I have an interesting place to go to, if you want to.

GRAYSON: Yes.

MACFARLANE: We can walk to it, if you're up to that. [. . .]

[END OF AUDIO, FILE #1.1]

GRAYSON: Looks like we're ready to go again after a very pleasant lunch at [Café] Eccell. [. . .] Okay, very good. One of the things that occurred to me while we were having lunch is the fact that you were a senior faculty member, but you're still here doing research. A lot of institutions have this, you know, retirement, or push you out the door, or tell you it's time to hang it up, and beat it. Is A&M pretty generous that way?

MACFARLANE: I think it's very much dependent on the individual, because I'm worried about this. Right now, federally, they can't fire me because of my age, but they can make life difficult for you if you're not producing. And by not producing, it means that you don't have research grants, and that you're not teaching an undergraduate course. The two of us that are at this advanced age both are aware of this and are both very active in our research and our teaching. So any time, if there is an attempt to suggest I leave, I said put it in writing. And so that's my game plan.

So far, Dave [David H.] Russell has been kind to me. I think he's kind of ambivalent. I know he was the one that [offered me] distinguished professor. A lot of things that he did were kind of unusual. We had a collaboration with Johns Hopkins [University] that involved them preparing some cell cultures for which we were to send them some samples of our HDL, and the timing was critical. And I don't know if I should even tell you this, because it's not really relevant to the . . . but I'll give you the example of what happens around here all the time. At that time, they decided—there was too much mold in the buildings and there was a mold remediation scheduled for the day that we were supposed to be preparing the samples. So, that meant that we couldn't send the samples, even though the cell culture was ready. So, I mentioned this to the dean—there was some sort of a social event. I said, "I got a problem here." He says, "I don't see what the problem is. I'll just tell the remediation to delay a week, that's no big deal."

Well, Dave got wind of this, and shit hit the fan. "You're doing this behind my back, and it's a bust! They have a schedule, and they have to do the remediation that day, period." And he was pissed off at the dean. The dean got all nervous about it, because he—the dean—did not

want to upset Dave. So that's what happened. It was several thousand dollars' worth of cell cultures that went down the drain. So they had quite a fight about it. So is Dave really on my side? My bottom line right now is he is, you know, with some reservations. When I told him I was funded again: "Oh, really?" Instead of saying, "Wow, congratulations;" "Oh, really," because I think that's probably, the natural exit is I lose my funding, and then it's . . .

GRAYSON: [Yes]. Well, that's kind of like the gold standard of whether you can hang in here. So now, Dave is the chair of the department of chemistry?

MACFARLANE: [Yes].

GRAYSON: So he's the guy that was sitting in that office over in 107?

MACFARLANE: Not really. He spends more time, I think, in the other building.

GRAYSON: [Yes], he does. But I mean that's his office over in Room 107.

MACFARLANE: Right.

GRAYSON: Well, in this school, it's probably a very important position. A lot of schools . . .

MACFARLANE: I can't read Dave very well, because I get two readings from him, one that is supporting me, and one that is neutral. The sooner that I leave, the sooner he can hand that position to someone younger. My feeling is that, as long as I'm productive and now I have a website, and it's a chat room for other faculty who want to get ideas for teaching, he went along with that. So I'm just kind of being neutral on it, and just kind of moving with the flow, and keep my antennae up.

GRAYSON: I mean, you want to stay in this environment as long as you can? <**T: 05 min**>

MACFARLANE: That's right.

GRAYSON: [Yes]. Well, it's challenging intellectually and sounds like it's a lot of fun. I mean that's what it's really about in the final analysis.

MACFARLANE: [Yes]. But I think that it has a survival aspect, I have to be aware that I'm vulnerable. My other colleague, who is a couple years older than I am, feels the same way. You're talking about contributing to the department

GRAYSON: So the chemistry department's a bit far-flung, because the building here is with cardiovascular disease, was it? What's the title again?

MACFARLANE: Laboratory for Cardiovascular Chemistry.

GRAYSON: Cardiovascular chemistry. [Yes], we got the chemistry in there. [Yes]. Well, that's an interesting mixture of concepts. Okay. Well, I just wanted to . . . because I know this is an issue that John Fenn faced at Yale—issues with them trying to . . . well, I guess, they did eventually kind of push him out.

MACFARLANE: [Yes], right.

GRAYSON: But it's interesting. Winning Nobel Prize [Chemistry, 2002] gets you back in, [Yes]. Let's see . . . how we doing with this little outline. What've we gone through? Well, now, you do have a couple of patents. How does the school treat patents? Is that . . .

MACFARLANE: [Yes], very well.

GRAYSON: Very well.

MACFARLANE: They are kind of novices at it. The first time this became an issue was when we had a collaboration with Johns Hopkins. And they were looking at the cord blood of newborns and wanted to know if our method would distinguish between newborns born smaller than they should be and newborns born normal size. And their data showed that if you're prematurely born, your chances of getting premature heart disease is a lot higher. They wanted to know why. Is there anything in their blood, when they're newborns—the cord blood—that would distinguish that. We agreed to run some samples for them. We didn't know which is which and our results was that there are two populations, one population is this, one population is that, and they were dumbfounded, because we separated the smaller than they should be from the normal.

GRAYSON: This is looking at cord blood?

MACFARLANE: Cord blood.

GRAYSON: Analyzing it by what method . . .

MACFARLANE: Lipoprotein method . . .

GRAYSON: Which is the . . . okay.

MACFARLANE: Density profile. And so there's a bump in the HDL that was there for some but not for the others. That bump was due to being born more small than they should be. They get all excited about that, and we have actually been able to get a *JAMA* [*Journal of the American Medical Association*] article on it.[15] That was a bit of a tour de force, because they never before—this was based on something they initiated. We had to put in language that an MD would understand. So that was a bit of a challenge. We did it, and it got published. And so then I said to our collaborator, I said, "This looks like something that should be patented." "Well, we've been thinking about that." Nothing happened. Nothing happened. Then I mentioned it to our technology transfer people, and they said, "Well let's look into it." He emailed back and said they've already filed a patent and your name is not on it. They contacted the patent office at Hopkins and they said, "Sue us."

GRAYSON: Oh, wow.

MACFARLANE: I tell you the story because it means that our office, technology transfer, was weak. And so they admit this. They just don't have the power that Johns Hopkins has. So then more issues came up and then there was another discovery that we made for which we got a patent. Then one of our alumni ran a business that was pretty good. He sold it, and he had some money that he wanted to invest, so he came to department and says, "Is there anything ongoing that looks like it might have some financial traction?" And he <**T: 10 min**> mentioned our project.

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[15] P. O. Kwiterovich, S. L. Cockrill, D. G. Virgil, E. S. Garrett, J. Otvos, C. Knight-Gibson, P. Alaupovic, T. Forte, L. Zhang, Z. N. Farwig, and R. D. Macfarlane, "A Large High-density Lipoprotein Enriched in Apolipoprotein C1: A Novel Biochemical Marker in Infants of Lower Birth Weight and Younger Gestational Age." *JAMA* 293, no. 15 (2005): 1891-1899.